

Assignment: Binary Search II

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Easy (6 Questions)

1. Sqrt(x)

Find the integer square root of a number.

- LeetCode: <https://leetcode.com/problems/sqrtx/>
 - GFG: <https://www.geeksforgeeks.org/problems/square-root/1>
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2. Arranging Coins

Find the maximum number of complete rows.

- LeetCode: <https://leetcode.com/problems/arranging-coins/>
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3. Koko Eating Bananas

Find the minimum eating speed.

- LeetCode: <https://leetcode.com/problems/koko-eating-bananas/>
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4. Minimum Days to Make m Bouquets

Find the minimum day required.

- LeetCode: <https://leetcode.com/problems/minimum-number-of-days-to-make-m-bouquets/>
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5. Capacity To Ship Packages Within D Days

Find the minimum ship capacity.

- LeetCode: <https://leetcode.com/problems/capacity-to-ship-packages-within-d-days/>
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6. Allocate Minimum Number of Pages

Classic Binary Search on Answer problem.

- GFG: <https://www.geeksforgeeks.org/problems/allocate-minimum-number-of-pages0937/1>
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Medium (7 Questions)

1. Split Array Largest Sum

Minimize the largest subarray sum.

- LeetCode: <https://leetcode.com/problems/split-array-largest-sum/>
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2. Magnetic Force Between Two Balls

Maximize the minimum distance.

- LeetCode: <https://leetcode.com/problems/magnetic-force-between-two-balls/>
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3. Aggressive Cows

Maximize the minimum distance between cows.

- GFG: <https://www.geeksforgeeks.org/problems/aggressive-cows/1>
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4. Painter's Partition Problem

Minimize the maximum time taken.

- GFG: <https://www.geeksforgeeks.org/problems/the-painters-partition-problem1535/1>
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5. Minimize Max Distance to Gas Station

Minimize the maximum distance after adding stations.

- GFG: <https://www.geeksforgeeks.org/problems/minimize-max-distance-to-gas-station/1>
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6. Find the Smallest Divisor Given a Threshold

Find the minimum divisor satisfying the threshold.

- LeetCode: <https://leetcode.com/problems/find-the-smallest-divisor-given-a-threshold/>
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7. Minimized Maximum of Products Distributed to Any Store

Distribute products while minimizing the maximum per store.

- LeetCode: <https://leetcode.com/problems/minimized-maximum-of-products-distributed-to-any-store/>
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Hard (6 Questions)

1. K-th Smallest Pair Distance

Binary search on distance.

- LeetCode: <https://leetcode.com/problems/find-k-th-smallest-pair-distance/>
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2. Maximum Value at a Given Index in a Bounded Array

Find the maximum feasible value.

- LeetCode: <https://leetcode.com/problems/maximum-value-at-a-given-index-in-a-bounded-array/>
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3. Minimum Limit of Balls in a Bag

Minimize the maximum balls in any bag.

- LeetCode: <https://leetcode.com/problems/minimum-limit-of-balls-in-a-bag/>
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4. Maximum Candies Allocated to K Children

Maximize candies each child receives.

- LeetCode: <https://leetcode.com/problems/maximum-candies-allocated-to-k-children/>
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5. K-th Smallest Number in Multiplication Table

Binary search on value space.

- LeetCode: <https://leetcode.com/problems/kth-smallest-number-in-multiplication-table/>
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6. Nth Magical Number

Find the nth number divisible by either of two numbers.

- LeetCode: <https://leetcode.com/problems/nth-magical-number/>